

Block Transfers, Safe Screening, and Linear Convergence for the Triangle Algorithm Family of SVM Solvers

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Abstract

The Triangle Algorithm (TA) solves hard-margin support vector machine training as a convex-hull distance problem with a certified duality gap. We extend the enhanced TA of Gupta and Kalantari to the full SVM family — L_2 and L_1 (hinge/ ν -SVM) soft margins and kernels — through exact reductions that reuse the algorithm’s dot-product caches, and we introduce three algorithmic improvements: *block weight transfers* that amortise each $O(n)$ scan over k MDM-style transfers via one exact line search along an aggregated direction; *gap-certified safe screening* that provably removes non-support points while preserving all certificates; and a *guarded step rule* with an away-step fallback. Our main theoretical result is global linear convergence of the guarded block algorithm with explicit constants and no combinatorial swap-step bounds, obtained by analysing the method on the Minkowski-difference polytope and inheriting the pyramidal-width machinery of Lacoste-Julien and Jaggi. Empirically, the optimised solver is the fastest method to a certified solution on six of seven benchmark classes against LIBSVM, LIBLINEAR, and SMO baselines, and every inequality in the analysis is verified numerically at machine precision.

1 Introduction

2 Background and setting

Let $V = \{v_1, \dots, v_n\}$, $W = \{w_1, \dots, w_m\} \subset \mathbb{R}^d$ and

$$\min_{p \in \text{conv}(V), q \in \text{conv}(W)} f(p, q) = \frac{1}{2} \|p - q\|^2. \quad (1)$$

With $z = p - q$ and the Minkowski-difference polytope $Z = \text{conv}(V) - \text{conv}(W) = \text{conv}\{v_i - w_j\}$, (1) is the minimum-norm-point problem over Z , whose objective is 1-strongly convex and 1-smooth. The Triangle Algorithm [1, 2] maintains an iterate pair with convex weights (α, β) and a certified interval $[\text{LB}, \text{UB}]$ from parallel supporting hyperplanes; it stops when $\text{UB} - \text{LB} \leq \varepsilon \text{UB}$.

2.1 Exact reductions

L_2 soft margin: $K \mapsto K + I/C$ via slack coordinates $\tilde{x}_i = [x_i; e_i/\sqrt{C}]$, never materialised (sparse diagonal corrections to the caches). ν -SVM: reduced convex hulls $R(V, \mu) = \{\sum a_i v_i : \sum a_i = 1, 0 \leq a_i \leq \mu\}$, $\mu = 2/(\nu\ell)$; the extreme-point oracle becomes a capped top- k blend. Kernels: all quantities are inner products; iterates live in weight space.

*Draft skeleton generated from the replication and extension study; see `docs/block_convergence.md` and `REPORT.md` in the accompanying repository.

3 The guarded block-transfer algorithm

Algorithm 1 Guarded block transfer with away fallback (one side, one iteration)

- 1: scores $s_i \leftarrow \langle g, v_i \rangle$; top- k receivers, worst- k active donors, paired best-with-worst; pair 1 is the MDM pair (r^*, u^*)
 - 2: per-pair steps $\gamma_j \leftarrow \min(\text{gap}_j/s_j, c_j)$
 - 3: **if** pair 1 uncapped **then**
 - 4: take the better of the block step (exact line search along $d = \sum_j \gamma_j d_j$) and the single MDM step $\triangleright$ guard
 - 5: **else**
 - 6: **if** away step on u^* is boundary-clipped **then**
 - 7: take it $\triangleright$ drop step: $\alpha_{u^*} \rightarrow 0$
 - 8: **else**
 - 9: take the best of $\{\text{block}, \text{MDM}, \text{away}(u^*), \text{toward}(r^*)\}$
 - 10: **end if**
 - 11: **end if**
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4 Analysis

All proofs are in the repository note `docs/block_convergence.md` and will be inlined here.

Lemma 1 (Gap separability). *With the product representation $\lambda_{ij} = \alpha_i \beta_j$ over Z , $g^{\text{PW}} = g_V^{\text{PW}} + g_W^{\text{PW}}$, and likewise for the Frank–Wolfe and away gaps; side transfers are pairwise directions of Z .*

Lemma 2 (Block gain). *With $S = \langle g, d \rangle$, $D = \|d\|^2$: $S \geq \sum_j \gamma_j^2 s_j$, $D \leq kS$, and the realised block gain satisfies $\Delta_B \geq S/(2k) \geq \Delta_{\text{MDM}}/(2k)$.*

Theorem 1 (Global linear convergence). *For any block size $k \geq 1$, the guarded block-transfer Triangle Algorithm with away fallback satisfies, with $\delta = \text{PW}(Z)$, $M = \text{diam}(Z)$, $\rho = c\delta^2/M^4$ for an explicit absolute constant c : every non-drop iteration contracts, $h_{t+1} \leq (1 - \rho)h_t$; drop iterations number at most $kT_{\text{good}} + s_0$; hence $h_T \leq h_0(1 - \rho)^{(T-s_0)/(k+2)}$.*

Proposition 1 (Near-orthogonal speedup). *If $|\langle d_i, d_j \rangle| \leq \eta\sqrt{s_i s_j}$ for $i \neq j$, the interior block gain is at least $(\sum_j \Delta_j)/(1 + \eta(k - 1))$.*

Theorem 2 (Screening activation). *Let $\delta^* > 0$, let τ be the minimal nonzero optimal-face score margin, D the larger class diameter and $R = \max_{z \in Z} \|z\|$. The certified gap equals the Frank–Wolfe gap over Z divided by UB , and once $h_t \leq \tau^4 \delta^{*2}/(2592 R^4 D^4)$ every zero-weight point outside the optimal faces is screened at the next round, permanently; under the linear rate this holds for all $t \geq T^* = s_0 + \frac{k+2}{\rho} \log(2592 R^4 D^4 h_0/(\tau^4 \delta^{*2}))$, after which every scan costs $O(|F| + |\text{supp}|)$ instead of $O(n + m)$. The τ -dependence is information-theoretically necessary.*

Theorem 3 (Safe screening). *With $r = \sqrt{\text{UB}^2 - \text{LB}^2}$, any zero-weight point with $(\langle h, v_i \rangle - \langle h, v_{\min} \rangle) > r\|v_i - v_{\min}\|$ is outside the optimal support; removing all such points preserves the optimum and the validity of all subsequent lower bounds.*

5 Experiments

6 Related work

7 Conclusion

References

- [1] B. Kalantari. A characterization theorem and an algorithm for a convex hull problem. *Annals of Operations Research*, 2015.
- [2] M. Gupta, B. Kalantari. An enhanced triangle algorithm for large-scale support vector machine optimization. *JIDMIS*, 3(9s), 2026.
- [3] S. Lacoste-Julien, M. Jaggi. On the global linear convergence of Frank–Wolfe optimization variants. *NeurIPS*, 2015.
- [4] B. F. Mitchell, V. F. Demyanov, V. N. Malozemov. Finding the point of a polyhedron closest to the origin. *SIAM J. Control*, 12(1), 1974.
- [5] J. Guelat, P. Marcotte. Some comments on Wolfe’s ‘away step’. *Mathematical Programming*, 35, 1986.
- [6] L. El Ghaoui, V. Viallon, T. Rabbani. Safe feature elimination for the lasso and sparse supervised learning problems. *Pacific J. Optimization*, 2012.